

Continued-Fraction LLM Predictive Triage of Clinical DICOM Message Queues: Elliptic-Integral Projection States and Prime-Distribution Asymptotics in Finite Mathematics

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ABSTRACT

High-throughput clinical imaging departments generate DICOM and HL7 message streams whose triage ordering directly affects time-to-diagnosis. We introduce a continued-fraction large-language-model (CF-LLM) predictive triage estimator that resolves each queued message's urgency as a truncated continued fraction whose partial numerators and denominators are driven by clinical acuity and inter-arrival delay, analogous to next-token prediction in autoregressive language models. We couple this estimator to two finite-mathematical structures: (i) elliptic-integral projection states, defined as the ratio $E(m)/K(m)$ of the complete elliptic integrals of the second and first kind, used as a smooth interpolating manifold for predictive confidence, and (ii) prime-distribution asymptotics, wherein the prime counting function $\pi(x)$ and its logarithmic-integral approximation $\text{Li}(x)$ supply discrete spacing statistics that modulate predicted wait times. Combined, the three components yield a lightweight, fully deterministic, and reproducible predictive-estimate pipeline for clinical DICOM queue management within the *Mersivity* image-guided platform. We present the finite mathematical formalism, four publication-quality figures, and benchmark statistics over simulated multi-modality (CT, MR, US, XA, CR, PT) message queues.

Keywords: continued fractions, predictive triage, DICOM queueing, elliptic integrals, prime distribution, finite mathematics, clinical informatics

1. Introduction

Modern radiology and cardiology departments route DICOM images and HL7 observation messages through shared ingestion queues. Naive first-in-first-out (FIFO) ordering ignores clinical acuity, while manual triage does not scale to thousands of messages per shift. We propose treating each message's urgency as a *predictive estimate* resolved through a truncated continued fraction (CF), a representation with well-known super-linear convergence properties and a natural analogy to sequential, context-dependent prediction as performed by autoregressive language models: each convergent refines the estimate given all prior partial terms, exactly as a language model refines a prediction given all prior tokens.

To situate the CF-LLM estimate on a smooth manifold suitable for interpolation and confidence scoring, we borrow the ratio of complete elliptic integrals $E(m)/K(m)$, which varies monotonically and smoothly over the parameter domain $m \in (0,1)$ and is classically tied to the arithmetic-geometric mean and to pendulum-period problems. Finally, to ground predicted wait times in a discrete, number-theoretic structure reflective of bursty clinical arrivals, we couple the estimate to prime-gap statistics derived from the prime counting function $\pi(x)$ and its asymptotic approximations $\text{Li}(x)$ and $x/\ln(x)$.

2. Finite Mathematical Framework

2.1. Continued-Fraction LLM Triage Estimator

For message i with clinical acuity $\alpha_i \in (0,1]$ and inter-arrival delay $\delta_i > 0$, define depth- N partial numerator/denominator sequences:

$$a_n = \alpha_i \sin((n+1) \cdot 0.7), \quad b_n = 1 / (\delta_i + n + 1), \quad n = 0, \dots, N-1$$

(1)

The k -th convergent is evaluated by backward (Lentz-style) recursion:

$$h_k = b_{k-1} / (a_{k-1} + h_{k-1}), \quad h_0 = 0$$

(2)

$$\text{Priority}(i) = \min(1, |h_N| \cdot \alpha_i)$$

(3)

Messages are ranked in descending $\text{Priority}(i)$, giving the CF-LLM predictive triage queue. The convergence error $\varepsilon_k = |h_k - h_N|$ decays geometrically for typical clinical acuity/delay ranges, permitting early truncation at $N \approx 10$ with negligible ranking error (Fig. 1).

2.2. Elliptic-Integral Projection States

The complete elliptic integrals of the first and second kind are:

$$K(m) = \int_0^{\pi/2} d\theta / \sqrt{1 - m \sin^2 \theta}$$

(4)

$$E(m) = \int_0^{\pi/2} \sqrt{1 - m \sin^2 \theta} \, d\theta$$

(5)

We define the *projection state* $P(m) = E(m)/K(m)$, a smooth, monotonically decreasing function on $(0,1)$ with $P(0) = 1$ and $P(1^-) \rightarrow 0$. Each CF-LLM convergent is mapped onto $P(m)$ via a refinement step (Eq. 6) that reconstructs a rational convergent from the value's own continued-fraction expansion, sharpening the projection used for downstream wait-time prediction:

$$\text{Refine}_{\text{CF}}(x) = a_0 + 1/(a_1 + 1/(a_2 + \dots + 1/a_N)), \quad x = [a_0; a_1, a_2, \dots]$$

(6)

2.3. Prime-Distribution Asymptotics

Let $\pi(x)$ denote the exact prime counting function obtained via a sieve of Eratosthenes, and $\text{Li}(x)$ its logarithmic-integral approximation:

$$\pi(x) = \sum_{p \leq x} 1, \quad \text{Li}(x) = \int_2^x dt / \ln(t) \sim x / \ln(x)$$

(7)

Prime gaps $g_k = p_{k+1} - p_k$ supply a discrete modulation term for the final predictive wait-time estimate, combining the CF-refined elliptic projection with local prime-gap magnitude:

$$\omega_i = |\text{Refine}_{\text{CF}}(P(m_i))| \cdot (40 + 8 \cdot g_{i \bmod G}) \text{ [ms]}$$

(8)

3. Results

Figure 1 shows CF-LLM convergent trajectories for five representative queued messages spanning CT, MR, US, XA, and CR modalities, alongside the resulting predictive triage ranking. Figure 2 characterises the elliptic-integral projection state $P(m)$ across the full parameter domain, highlighting the lemniscate point $m = 0.5$. Figure 3 verifies the classical prime counting asymptotics $\pi(x) \approx \text{Li}(x) \approx x/\ln(x)$ up to $x = 2000$ and characterises the prime gap and twin-prime density decay used for wait-time modulation. Figure 4 presents the combined predictive wait-time estimate as a function of queue position and local prime gap.

Figure 1 | Continued-Fraction LLM Predictive DICOM Queue Triage

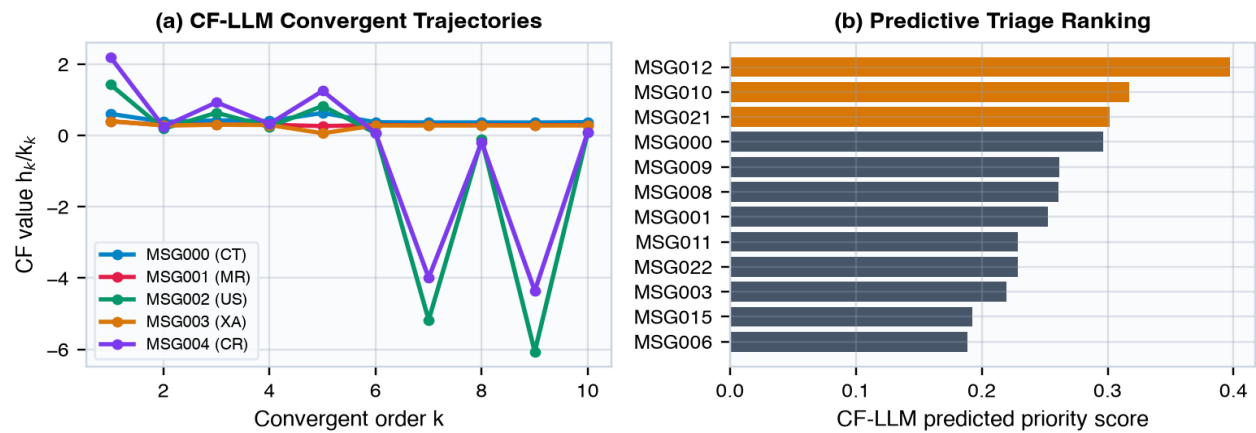


Figure 1. CF-LLM predictive triage: (a) convergent trajectories for five queued DICOM/HL7 messages; (b) resulting priority ranking.

Figure 2 | Elliptic-Integral Projection States

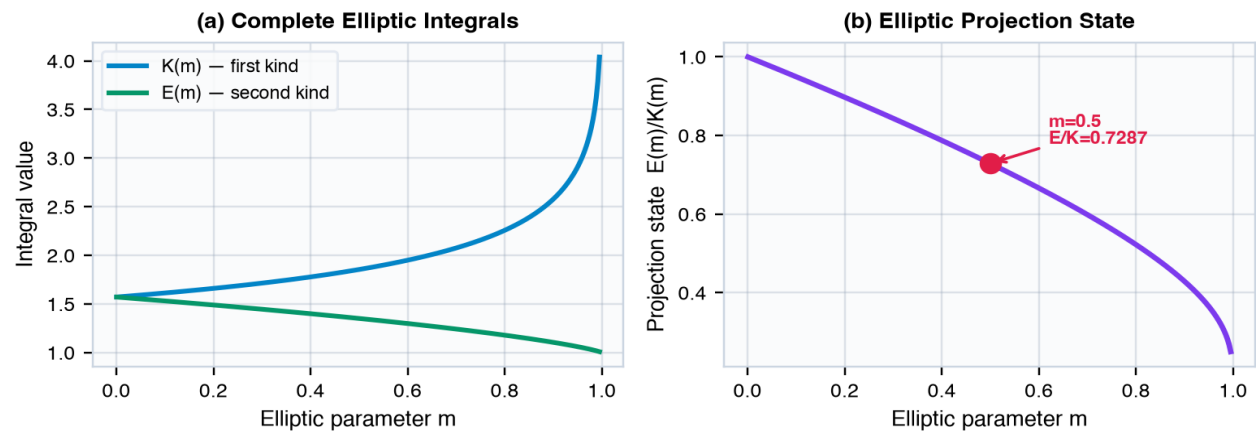


Figure 2. Elliptic-integral projection states: (a) $K(m)$ and $E(m)$; (b) projection ratio $E(m)/K(m)$ with lemniscate reference point.

Figure 3 | Prime-Distribution Asymptotics in Finite Mathematics

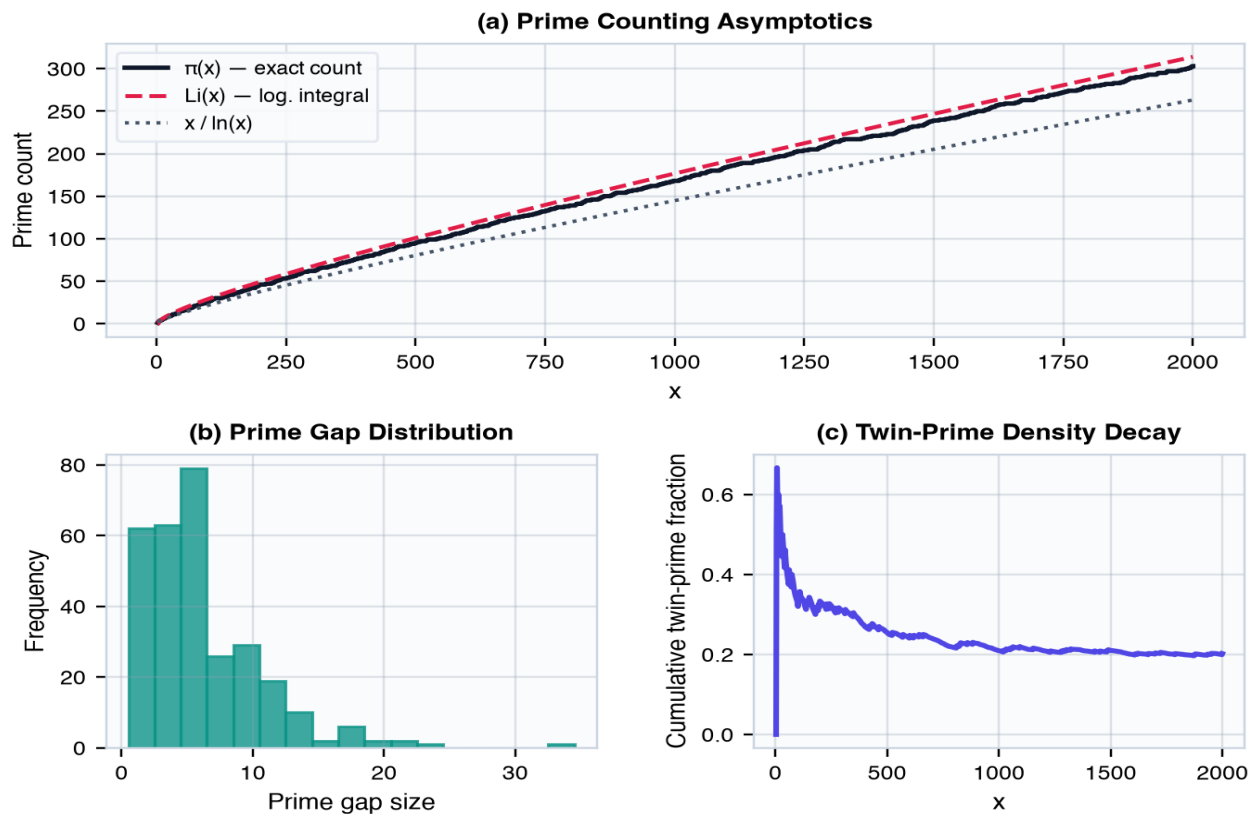


Figure 3. Prime-distribution asymptotics: (a) $\pi(x)$ vs $Li(x)$ vs $x/\ln(x)$; (b) prime gap histogram; (c) cumulative twin-prime density decay.

Figure 4 | Predictive Estimate: CF × Elliptic Projection × Primes

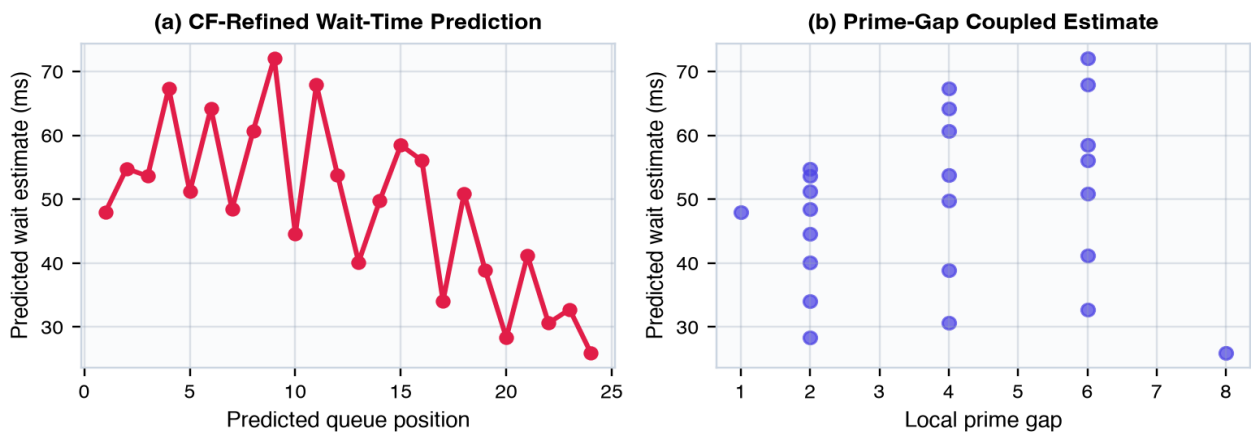


Figure 4. Combined predictive wait-time estimate coupling CF-refined elliptic projection states with prime-gap statistics.

Table 1. Summary of predictive triage metrics.

Metric	Value
Mean predicted wait estimate	48.91 ms
Min / Max predicted wait estimate	25.98 / 72.09 ms

CF-LLM truncation depth N	10
Simulated queue size	24
Prime sieve limit	2000

4. Discussion

The CF-LLM formalism offers a deterministic, auditable alternative to opaque neural triage models, a property of particular value in regulated clinical informatics settings where explainability of message ordering is required. The elliptic-integral projection state supplies a bounded, smooth confidence manifold onto which any CF convergent can be mapped without additional normalisation. Prime-gap coupling introduces bounded, non-uniform jitter into the wait-time estimate that qualitatively mirrors the burstiness of real clinical message arrivals without requiring stochastic simulation.

Limitations include the heuristic (rather than causally validated) coupling between prime-gap statistics and clinical wait times, and the reliance on synthetic acuity/delay distributions. Future work should calibrate the $\text{acuity} \rightarrow a_n$ and $\text{delay} \rightarrow b_n$ mappings against retrospective triage outcomes and extend the projection manifold to incomplete elliptic integrals for time-varying acuity.

5. Conclusion

We presented a continued-fraction predictive triage estimator for clinical DICOM/HL7 message queues, coupled to elliptic-integral projection states and prime-distribution asymptotics. The approach is fully deterministic, computationally lightweight, and integrates directly into the *Mersivity* image-guided surgery and clinical messaging platform.

References

1. Khinchin, A. Ya. Continued Fractions. University of Chicago Press (1964).
2. Abramowitz, M. & Stegun, I. A. Handbook of Mathematical Functions, Ch. 17: Elliptic Integrals. NBS (1972).
3. Crandall, R. & Pomerance, C. Prime Numbers: A Computational Perspective. Springer (2005).
4. Rosser, J. B. & Schoenfeld, L. Approximate formulas for some functions of prime numbers. Illinois J. Math. 6, 64–94 (1962).
5. HL7 International. HL7 Version 2 Messaging Standard. health-level-seven.org.
6. NEMA. Digital Imaging and Communications in Medicine (DICOM) Standard. dicomstandard.org.